

Customer Sphere Design

HawkinsTwin Customer 360 — one customer, one schema, data that stays yours

PREPARED FOR

Tim Hawkins · Luke Weatherbie

Hawkins Truck Mart's customer records live in a dealer management system, a call platform, and the systems Craig Allen runs day to day. This design resolves all of them onto one customer, in one combined schema, with one key. The description of that schema is published and portable. The data itself stays inside Peterbilt Atlantic's own Azure tenant and never leaves it. That split is the whole design, and it is what puts the dealership — not a vendor — in the reference-model position for its industry.

Status — Draft for review. No dealership data has moved.

Agreement holder Tim Hawkins · **20 Group access** Luke Weatherbie · **Systems register** Craig Allen

Source of record [customer-sphere/CUSTOMER-SPHERE-DESIGN.md](#)

Issued 2026-08-02T02:44:21Z · **Key ID** EgD-KEY-2026-07 · **Pages** 10

SHA-256 ff7433ff79d83b96aa3a5f4b4010b54f32718d90d2bd9d42523b3472bf74299f



Programme: EVEglyphDesign — Peterbilt Atlantic Digital Twin **Direction record:** 2026-08-01 — Customer 360 and the reference-model position **Agreement holder:** Tim Hawkins · 20 **Group access:** Luke Weatherbie · **Systems register:** Craig Allen **Constraint:** the live CDK dealer management system is never written to. **Status:** draft. No dealership data has moved.

1. Why a sphere and not a pipeline

A pipeline has a direction. Data leaves a system, is transformed, and lands somewhere downstream where someone looks at it. That shape is adequate for reporting and inadequate for a customer, because a customer is not downstream of anything. The same fleet owner is simultaneously a sales record in CDK, a parts account, a service history, an open warranty claim, six unreturned voicemails, a finance file, and a name Craig's marketing systems know under a different spelling.

The customer sphere is the surface on which all of those faces resolve to one entity. The integrator's job, as it has been stated across this programme, is to hold economics, operations, and relationship in a single event horizon and let the sphere emerge from the triangulation. This document is that principle applied to one dealership's customers.

Practically: **one combined schema, one customer key, many faces.** No face is authoritative over the others. Each face carries its own source, its own timestamp, and its own confidence, and the sphere records disagreement rather than averaging it away.



2. The two-body split

This is the governing constraint of the design and every other decision is subordinate to it.

	Metadata repository	Business data store
What	Schema, field dictionary, identity rules, lineage, model definitions, DDL, validators, tests, change history	Actual customer, sales, service, parts, finance and call records
Where	GitHub — HawkinsTwin Customer 360 metadata repository	Postgres, inside Hawkins's own Azure tenant
Custody	EVEglyphDesign authors, Hawkins owns; readable by any party that needs to understand the model	Hawkins alone. EVEglyphDesign holds no copy
Contains a customer record?	Never	Yes, all of them
Portable?	Yes, deliberately — it is the reference model	Yes, but it moves only on Hawkins's instruction

The split answers the question every dealership principal eventually asks, which is not *what does it do* but *who ends up holding my data*. The answer here is that the description of the data is public and forkable, and the data itself never leaves the tenant Hawkins already pays for and already controls. A vendor cannot hold the dealership hostage with a schema it refuses to document, and cannot hold it hostage with data it refuses to hand back, because it never had either.

A secondary consequence matters for the commercial case in §7: because the metadata is complete enough to rebuild the model from scratch, the model is separable from the data. That is what makes it sellable as a reference model without selling anyone's customers.



3. Sources into the sphere

Three source families, one schema.

3a. CDK — customer and sales

The dealer management system is the spine. Customer master, sales transactions, deals and F&I, repair orders, parts invoices, warranty claims, and the general ledger tie-out. Acquisition follows the existing two-lane design in the [reconciliation contract](#): the restored-backup lane for completeness, the API and export lane for currency, both landing in the same Postgres and reconciled on demand rather than silently merged. The field-level model — 21 objects and 443 confidence-tagged fields — is the starting dictionary for the customer and sales portion of the combined schema, not a separate artefact.

3b. Call history — the relationship record

The TELUS call platform supplies inbound and outbound call events, direction, duration, extension, disposition and voicemail state. This is the only source in the sphere that records what the dealership *did about* a customer rather than what it *sold* one. Its three standing cohorts — voicemail not returned within two hours, repeat caller without callback, and never reached — are the sphere's first behavioural signals. Ingestion runs under a Peterbilt-owned service identity so custody and audit trail stay with the company. Where call direction or extension identity is missing, the record contributes to cohort counts but is never used to score an individual.

3c. Craig's register — the operating surface

The sixteen systems catalogued by Craig Allen are treated as first-class sources, not as context. Marketing and lead platforms, inventory and ePortal surfaces, scheduling, and whatever else the register holds each contribute a face to the customer. Craig's register is the authority on what exists; this design's obligation is only to give each entry a defined landing in the schema, an owner, an access path, and a confidence tag — or an explicit note that it is deferred.

Open item. The source inventory is not closed until Craig confirms the register is complete and current, and until each system has a stated access route — API, export, or manual. Systems without an access route are recorded as known-but-unreachable rather than quietly dropped.



4. Identity — how a customer becomes one customer

Identity resolution is the hardest part of the sphere and the part most likely to be got wrong quietly, so it is specified before any model is built.

- **A durable internal key.** Every resolved entity carries a `customer_sphere_id` minted inside the sphere. No source system's primary key is promoted to the master key, because every one of them is a key for its own purposes and will eventually be reused, merged, or retired by its vendor.
- **Source keys are retained, never overwritten.** The CDK customer number, the phone identifier, and each of Craig's system keys are all kept against the resolved entity. Losing a source key destroys the audit path back to the record it came from.
- **Deterministic before probabilistic.** Exact matches on CDK customer number, on a normalised phone number, and on VIN-to-owner links are applied first. Only what remains unmatched goes to fuzzy matching on name, address, and business name.
- **Match confidence is stored, not thresholded away.** Each link carries a score and the rule that produced it. Low-confidence links are held as *candidate* links and surface in a stewardship queue rather than being auto-merged.
- **Merges are reversible.** Every merge is an appended, timestamped, attributed event. An unmerge restores both sides. A destructive merge is a defect, not a design choice.
- **Person and organisation are distinct.** A fleet is not a driver. The sphere carries both, with an explicit affiliation edge between them, because dealership relationships are almost always one organisation to several people.
- **Vehicle is a third axis.** Customer, organisation, and vehicle form the three anchors; service and warranty history hang from the vehicle, not from the person who happened to book the appointment.



5. Schema shape

Five layers, each with a job, all in the same Postgres and all described in the metadata repository.

1. **Landing.** Source-native. Original field names, original widths, nothing coerced. Where a source column is wider than the SAP-shaped default, the source width wins and the widening is recorded explicitly — dealer data is never truncated to fit a model.
2. **Conformed.** Types normalised, phone and address standardised, currency and dates unified, source keys preserved. Still one row per source record.
3. **Identity.** The resolution layer of §4 — entity tables, link tables, candidate queue, merge event log.
4. **Sphere.** The combined customer model. One customer, all faces: commercial history, service history, parts, warranty, finance, contact and call behaviour, marketing presence, and the cohort flags derived from them.
5. **Marts.** Analytical structures for the 20 Group performance set, service and parts performance, and the decision surfaces. These reuse the SAP-shaped analytical model and the [data mart library](#) rather than inventing a Hawkins-only structure — reuse is what lets a component be lifted to another dealership later without a rewrite.

Every field at every layer carries source, extraction timestamp, and confidence tag. A number that reaches a decision surface can always be walked back to the record and the extraction that produced it.



6. Governance and the gates

- **Nothing moves before the agreement.** Tim Hawkins's executed agreement gates any real dealership data. Until then the sphere runs on synthetic and mock composite data.
- **Peer comparison waits on access.** 20 Group peer figures populate only once Luke has access; blank variance fields stay blank rather than being guessed.
- **The live DMS is never written to.** Unchanged, in every phase.
- **Service identities, not personal credentials.** Every source connection is a company-owned identity with an attributable audit trail.
- **Tenant identity is checked centrally.** Every pull resolves the authenticated account and aborts on mismatch, in the shared token path, so no individual workflow can skip it.
- **Personal data stays in the tenant.** Names, phone numbers, addresses, account numbers and VINs exist only in Azure Postgres. The metadata repository holds field definitions and synthetic examples. Any commit that would place a real record in GitHub is a durability and privacy defect and is treated as one.
- **Retention and erasure are schema features.** Because Canadian privacy obligations apply to a New Brunswick dealership, the sphere carries retention policy per source and an erasure path that resolves through the identity layer, so a deletion request can be executed once and propagate rather than being chased across sixteen systems by hand.



7. Why this makes Hawkins the reference model

The commercial argument is not that the dealership gets better reports. It is structural.

A dealership that owns a fully described customer model — schema public, data private, lineage complete, models running in its own tenant — is in a position no vendor-hosted dealership can occupy. Its numbers are defensible to PACCAR and to a 20 Group because the derivation is inspectable. Its data is portable because the description of it is complete. Its components can be scaled and sold to peers without the underlying business data ever being part of the transaction.

That is the reference-model position: the industry ends up measuring itself against a shape that Hawkins owns, and Hawkins carries none of the custody risk that would normally be the price of being the example. Components scale commercially; the reference model stays with the dealership.



8. Open questions

1. Is the sixteen-system register complete and current, and what is the access route for each entry?
2. Which Azure subscription and region hosts the Postgres, and who administers it on the Hawkins side?
3. Does CDK entitlement permit the Data Export Tool over SFTP/PGP plus Dealer Data Exchange, which is the preferred route, or is a per-data-type third-party path required?
4. What is the retention policy per source, and who signs it off?
5. Which of Craig's systems hold customer contact data that duplicates CDK, and which is authoritative when they disagree?



Provenance and control

Document

EVEglyphDesign — Hawkins Twin Customer Sphere Design, 10 pages.

Source of record

[customer-sphere/CUSTOMER-SPHERE-DESIGN.md](#) in the [Hawkins Twin Platform repository](#). The repository is authoritative; this PDF is a controlled copy of it at the timestamp below.

Direction record

[2026-08-01 — Customer 360 and the reference-model position](#), the dated exchange this design was written from.

Content hash (SHA-256 of the source Markdown)

ff7433ff79d83b96aa3a5f4b4010b54f32718d90d2bd9d42523b3472bf74299f

Key ID

EgD-KEY-2026-07

Issued

2026-08-02T02:44:21Z

Standing gates

No dealership data moves before Tim Hawkins's executed agreement. Peer comparison waits on Luke Weatherbie's 20 Group access. The live CDK dealer management system is never written to. Every surface runs on synthetic or mock data until those clear.

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Pour le bien-être du peuple.

